RESEARCH

EMLSCF-FCH: A Secure and Efficient Machine Learning-Based Framework for Fog Computing in Healthcare

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Abstract
The rapid integration of Internet of Things (IoT) technology in healthcare has led to the adoption of fog computing for real-time, delay-sensitive applications. However, ensuring secure communication and data integrity in fog computing-oriented healthcare remains a significant challenge. The abstract of EMLSCF-FCH is technically well-structured as it effectively addresses the healthcare domain while clearly identifying the problem of secure and efficient data management in fog computing. It outlines the objectives of leveraging machine learning to enhance security and performance, ensuring that the motivation behind the framework is evident. Sensitive medical data is at risk of privacy breaches and cyber threats, necessitating robust security frameworks for safeguarding patient information. This study proposes EMLSCF-FCH (Enhanced Machine Learning-based Secure Communication Framework in Fog Computing-oriented Healthcare), a novel approach to secure, optimize, and enhance disease prediction using ML. The framework leverages Enhanced Linear Regression (ELR), optimized by the Lotus Effect Optimization Algorithm (LEA), to improve prediction accuracy while minimizing computational overhead. Additionally, blockchain technology is incorporated to ensure data integrity and protect against unauthorized access. The proposed method achieves a computation cost by 97.60%, communication overhead by 96.58%, security robustness by 98.34%, a prediction accuracy of 99.05% and a real-time health monitoring of 97.36%.

Keywords Fog computing · Secure healthcare communication · Machine learning · Lotus effect optimization algorithm · Internet of Things (IoT) · Smart healthcare · Blockchain security · Disease prediction · Enhanced linear regression

1 Introduction

Through remote diagnosis, personalized treatment plans, and continuous patient monitoring, the IoT has transformed healthcare [1]. The explosion of healthcare data is generated by the explosion of connected medical devices, requiring efficient computer resources to process and analyse [2]. However, cloud computing, the traditional underpinning for healthcare applications driven by the IoT, suffers from latency issues, bandwidth constraints, and potential security vulnerabilities [3]. Fog computing represents a new approach that will bridge these limitations through the proximity of computing power to the IoT devices [4]. Thereby ensuring processing of data

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